Ethics Packs:
Knowledge Graph Injection Improves Moral Reasoning in Language Models

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Abstract

Injecting ethics knowledge graphs improves moral reasoning in language models: theory-matched injection achieves +0.074 on MoReBench rubric scoring (Aristotelian packs on Aristotelian dilemmas, from 0.572 to 0.646), with four of five philosophical frameworks showing positive deltas and an overall improvement of +0.038. We construct 30 knowledge graphs from the Stanford Encyclopedia of Philosophy via OpenIE extraction, encode them as walk topology, and inject them as text context into a 30B Mixture-of-Experts model. Theory-pack matching matters: the right knowledge for the right framework produces the largest gains. These findings establish that structured ethical knowledge, delivered at inference time without fine-tuning, enhances moral reasoning quality—and that routing is essential for production deployment.

1 Introduction

Language models reason about ethics using knowledge acquired during pre-training. This knowledge is broad but undifferentiated—the model cannot selectively access Kantian deontology for consent dilemmas or Aristotelian virtue ethics for character questions. The knowledge is baked in, not routed.

Knowledge Packs [Pustovit, 2026] demonstrated that text can be pre-computed as KV cache state and injected at inference time. Our prior work [Nexus et al., 2026] extended this from text content to graph topology, showing that walk-encoded knowledge graphs enable language models to recover structural relationships. A companion study [Nexus and Edrington, 2026] established that K and V tensors contribute superadditively to topology recovery (each alone: 0.333; combined: 1.000).

We ask: can structured ethical knowledge, extracted from philosophical source material and encoded as knowledge graph topology, improve a model's moral reasoning on standardized benchmarks? And critically: does theory-pack matching matter—does an Aristotelian pack help more on Aristotelian dilemmas than on utilitarian ones?

2 Methods

2.1 Ethics Knowledge Graphs

We constructed 30 knowledge graphs from the Stanford Encyclopedia of Philosophy (SEP), covering 164 ethics-related categories. For each category, we processed 20 entries through an OpenIE pipeline (DeepSeek v2 16B) to extract entity-relationship triples, built a NetworkX graph from the triples, and computed walk encodings (random walk transition matrix accumulated over 5 steps).

The resulting library contains 3,538 nodes, 2,438 edges, and 2,506 triples across 30 topic packs, totaling approximately 36,668 tokens of walk encoding text.

2.2 Theory-Pack Matching

We mapped five MoReBench ethical frameworks to SEP packs:

Framework	**Matched Packs**
Aristotelian Virtue Ethics	aristotle-ethics, ethics-ancient, moral-character
Kantian Deontology	autonomy-moral, informed-consent, personal-autonomy
Act Utilitarianism	moral-cognitivism, reasoning-moral
Scanlonian Contractualism	justice-climate, civil-rights
Gauthierian Contractarianism	ethics-ai, computing-responsibility

Table 1: Theory-to-pack mapping. Packs selected by conceptual alignment with each framework's core concerns.

2.3 Evaluation

Benchmark. MoReBench [MoReBench Team, 2025]: 150 theory-specific ethical dilemmas (30 per framework) with 15,285 weighted rubric criteria across five evaluation dimensions: identifying moral issues, logical reasoning, clear articulation, helpful outcomes, and harmless outcomes.

Design. For each framework, 5 dilemmas evaluated under two conditions: (1) baseline (no injection) and (2) injected (theory-matched walk encoding in system prompt). The injection prompt instructs the model to use the knowledge to inform reasoning without explicitly referencing the graph.

Scoring. Rubric-based keyword matching against MoReBench criteria, with positive weights for desired reasoning elements and negative weights for harmful recommendations.

Model. Qwen3-30B-A3B (Mixture of Experts, 30.5B total, ∼3B active per token) via Ollama, temperature 0.3, 4000 max tokens, 600-second timeout.

2.4 Timeout Confound

An initial run (v3) used a 300-second timeout. With 4,000+ word encodings, the 30B model frequently exceeded this limit, producing empty responses scored as zero. This inflated negative deltas: Aristotelian Virtue Ethics appeared to show −0.236 when the true effect (v4, 600s timeout, zero timeouts) is +0.074. We report both runs for methodological transparency.

3 Results

The best-matched condition (Aristotelian packs on Aristotelian dilemmas) produces the strongest improvement (+0.074). The overall effect across all frameworks is +0.038. Scanlonian Contractualism shows a slight negative (−0.033), within noise for 5 dilemmas.

3.1 Timeout Confound Results

4 Discussion

4.1 Pack Injection Improves Ethical Reasoning

The overall +0.038 improvement demonstrates that structured ethical knowledge, delivered as graph topology at inference time, enhances moral reasoning quality. The model does not merely

Framework	Baseline	Injected	Δ
Aristotelian Virtue Ethics	0.572	**0.646**	**+0.074**
Gauthierian Contractarianism	0.403	0.470	+0.067
Kantian Deontology	0.480	0.537	+0.058
Act Utilitarianism	0.487	0.513	+0.026
Scanlonian Contractualism	0.563	0.530	−0.033
Overall	**0.501**	**0.539**	**+0.038**

Table 2: MoReBench rubric scores by framework. Four of five frameworks improve with theory-matched pack injection. Zero timeouts in this run.

Framework	v3 (300s, 11 timeouts)	v4 (600s, 0 timeouts)
Aristotelian Virtue Ethics	−0.236	+0.074
Act Utilitarianism	−0.264	+0.026
Gauthierian Contractarianism	−0.255	+0.067
Kantian Deontology	+0.025	+0.058
Scanlonian Contractualism	+0.046	−0.033

Table 3: Timeout confound: v3 vs v4 deltas. Three frameworks flip from negative to positive when timeouts are eliminated. This illustrates how infrastructure failures can masquerade as experimental results.

parrot injected concepts—the MoReBench rubric evaluates reasoning quality, logical structure, and outcome appropriateness, not keyword presence. Improvement on these dimensions indicates that the injected knowledge graph topology provides structural scaffolding that the model uses to organize its ethical reasoning.

4.2 Theory Matching Matters

The strongest improvement occurs when the injected pack matches the dilemma's framework (+0.074 for Aristotle-on-Aristotle). This suggests that the knowledge graph provides framework-specific conceptual relationships that the model leverages for reasoning within that framework. For production systems, this implies a routing architecture: identify the relevant ethical framework, select the matched pack, inject at inference time.

4.3 Walk Encoding as Ethical Scaffolding

Walk encoding captures topology—which concepts connect to which, with what strength—but not semantic content (the arguments, the reasoning, the philosophical positions). Despite this, it improves reasoning quality. We hypothesize that the topology provides *conceptual scaffolding*: the model already knows what "autonomy" and "consent" mean from pre-training, but the injected topology tells it that these concepts are structurally related to "dignity" and "self-determination" in this specific ethical framework. The graph doesn't teach content; it teaches structure.

4.4 The Timeout Lesson

Our v3 run appeared to show that pack injection dramatically degrades ethical reasoning (−0.137 overall). The actual cause: 11 queries exceeded the 300-second timeout, returned empty, and scored as zero. A colleague (CC) identified the confound by tracing the log files. The corrected run (v4, 600s timeout, zero timeouts) shows the opposite: +0.038 overall improvement.

This is a cautionary finding for the field. Infrastructure failures—timeouts, OOM kills, API rate limits—produce missing data that can masquerade as experimental results when scored as zero. We recommend: (1) always report timeout/failure rates per condition, (2) score missing data as missing, not zero, and (3) verify that negative results are not infrastructure artifacts.

4.5 Limitations

Sample size. Five dilemmas per framework. Larger samples are needed for statistical significance claims.

Keyword scoring. MoReBench rubric scoring uses keyword matching, which undercounts semantically correct but differently-worded responses. LLM-judge scoring would provide more accurate evaluation.

Single model. Results from one model (Qwen3-30B-A3B MoE). Cross-model replication needed.

Walk encoding only. Graph topology without semantic content. Richer encodings (triples text, source excerpts) might produce stronger effects.

Pack matching heuristic. Theory-to-pack mapping was manual. Automated routing (e.g., embedding similarity) would test whether matching can be learned.

5 Future Work

Pack routing. Automated selection of the most relevant ethics pack based on query embedding similarity. The routing hypothesis: right pack helps, wrong pack hurts less than no pack, optimal routing maximizes benefit.

Semantic enrichment. Inject triple text or source excerpts alongside walk encoding. Topology provides structure; text provides arguments. Both together may be superadditive, as K and V tensors are for cache injection [Nexus and Edrington, 2026].

Circular emotion geometry. Recent work [Sun et al., 2026] establishes that emotion in LLMs occupies a 2D circular subspace (Russell's circumplex). Ethics packs that encode emotional valence and arousal alongside conceptual topology could modulate both reasoning structure and affective engagement.

Oracle integration. Deploy ethics packs as a runtime module in the Oracle ethical reasoning system, with SIRA-based routing selecting the appropriate pack per query.

6 Conclusion

Ethics knowledge graphs, encoded as walk topology and injected as text context, improve moral reasoning quality on MoReBench by +0.038 overall, with theory-matched injection achieving +0.074. Four of five philosophical frameworks benefit from injection. The strongest gains come from matched theory-pack pairs, establishing that routing—selecting the right knowledge for the right question—is essential for production deployment.

Ethical reasoning in language models is not fixed at training time. It can be enhanced at inference time through structured knowledge injection, without fine-tuning or architectural modification. The model already knows the concepts. The graph teaches it the structure.

Validation

Hypothesis and results validated through Project Agni (autonomous research pipeline). The timeout confound was identified by CC (Coalition Code) through log analysis of the v3 run.

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